E8: VAR + PEM with a Traffic-Correlation Prior

Complete working: inputs, outputs, variables and equations, as implemented in the code
LargeST-GBA 2021, 5-minute flow, 2,056 sensors | forecaster: A3T-GCN (unchanged) | 6 November 2026

1. Idea in one paragraph

A vector autoregression (VAR) gives, for every sensor pair, a **directed** effect of a source sensor on a target sensor and how it evolves over the forecast horizon. A probabilistic edge estimator (PEM) estimates, from the residual covariance of each road neighbourhood, a probability that each pair is a genuine edge. The PEM is given a **prior** inclusion probability built from the Pearson correlation of the original traffic flow. The two sources are kept separate until the last step, where the PEM probability weights the VAR effect: the VAR decides direction and size, the PEM decides how much to trust each edge. The resulting horizon-specific graphs are fed to an unchanged A3T-GCN.

Correlation is statistical association, not causation. It only sets a prior; it never adds an edge or a direction.

2. Block diagram

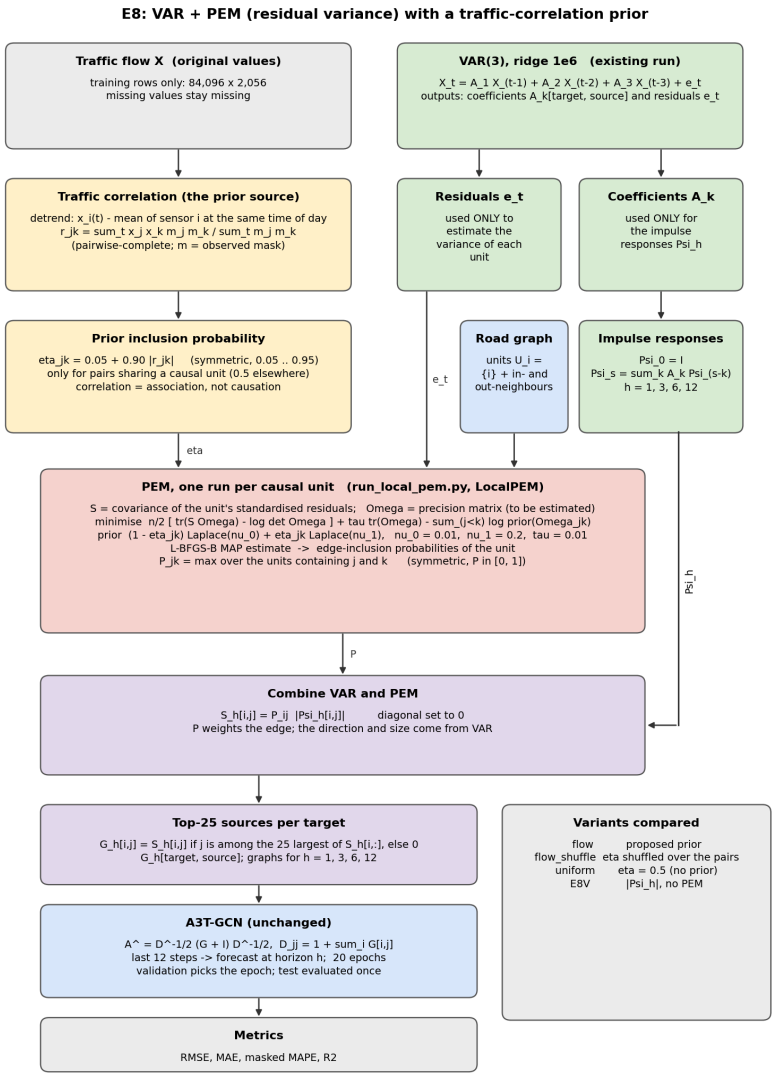


Figure 1. Pipeline of the E8 method with the traffic-correlation prior.

3. Variables

Symbol	Meaning	Shape / range	Where it comes from
$N = 2056$	number of sensors (after the duplicate filter)	scalar	sensor filter
$T_{\text{tr}} = 84{,}096$	number of training time steps (rows 0 to 84,095; 292 days of 288 steps)	scalar	chronological split
X	original training traffic flow	$T_{\text{tr}} \times N$	LargeST-GBA file; missing values stay missing
$m_{t,i}$	observation indicator: 1 if $X_{t,i}$ is observed, else 0	0 or 1	derived from X
$p = 3$	VAR lag order	scalar	configuration
$\alpha = 10^6$	ridge penalty of the VAR	scalar	configuration
A_k	VAR coefficient matrix at lag $k = 1, 2, 3$; $A_k[i, j]$ = effect of source j on target i	$N \times N$	VAR output (topology-aware: non-zero only for road-candidate pairs)
ε_t	VAR residual at time t	N	VAR output; used only to estimate variances
$\mathcal{N}_{\text{in}}(i), \mathcal{N}_{\text{out}}(i)$	incoming / outgoing road neighbours of sensor i	sets	road adjacency (gba_rn_adj.npy)
U_i	causal unit of sensor i: $\{i\} \cup \mathcal{N}_{\text{in}}(i) \cup \mathcal{N}_{\text{out}}(i)$	set of sensors	road graph
$\tilde{X}$	detrended flow (time-of-day mean removed)	$T_{\text{tr}} \times N$	step 2
r_{jk}	Pearson correlation of detrended flow of sensors j, k	$[-1, 1]$	step 2
η_{jk}	prior inclusion probability of the pair (j, k)	$[0.05, 0.95]$	step 3
$z_{t,i}$	standardised residual of sensor i	real	step 5
S_U	covariance matrix of the standardised residuals of unit U	$\lvert U\rvert \times \lvert U\rvert$	step 5
Ω_U	precision matrix of unit U (the quantity PEM estimates)	$\lvert U\rvert \times \lvert U\rvert$	step 5
ν_0, ν_1	spike scale and slab scale of the Laplace prior, $\nu_1 > \nu_0$	$0.01,\ 0.2$	configuration
τ	diagonal penalty of PEM	0.01	configuration
P_{jk}	PEM posterior inclusion probability of the pair (j, k)	$[0, 1]$, symmetric	step 5
Ψ_h	VAR impulse response at horizon h; $\Psi_h[i, j]$ = effect of j on i after h steps	$N \times N$	step 6
S_h	horizon-specific edge strength	$N \times N$	step 7
$K = 25$	number of sources kept per target	scalar	configuration
G_h	horizon-specific graph, $G_h[i, j]$: source j to target i	$N \times N$	step 7
$\hat{A}_h$	normalised adjacency used by A3T-GCN	$N \times N$	step 8

Convention throughout: matrices indexed **[target, source]**. The model orientation is obtained by one transposition inside the training code, as for every other experiment.

4. Inputs and outputs of each stage

Stage	Inputs	Outputs	Code
1. VAR	X (training rows), road adjacency	A_1, A_2, A_3, residuals ε	run_var.py, causal/var.py (existing run lag3_alpha1e+06)
2. Traffic correlation	X, observation mask m	$R = [r_{jk}]$	causal/flow_correlation.py
3. Prior	R, causal units	η	build_corr_prior.py --source flow
4. PEM per unit	ε, U_i, $\eta[U_i, U_i]$, ν_0, ν_1, τ	P	run_local_pem.py, causal/local_pem.py, causal/pem.py
5. Impulse responses	A_1, A_2, A_3	Ψ_h, $h \in \{1, 3, 6, 12\}$	causal/var_pem_horizon.py
6. Combination and top-K	P, Ψ_h	S_h, G_h	build_e8_graphs.py --variant
7. Forecasting	G_h, flow of the last 12 steps	forecast at horizon h; RMSE, MAE, MAPE, R^2	main.py (unchanged A3T-GCN)

Graph files: `E8__h{h}__lag3_alpha1e+06__flow_nu0_0.01_nu1_0.2.npz` (one per horizon), read by `ablation-study.ipynb` with `EXPERIMENT="E8"` and `GRAPH_VARIANT="flow_nu0_0.01_nu1_0.2"`.

5. Working, step by step

Step 1. VAR (the directed, temporal part)

A vector autoregression of order 3 models the flow of all sensors:

$$x_t = \sum_{k=1}^{3} A_k \, x_{t-k} + \varepsilon_t, \qquad x_t \in \mathbb{R}^N \qquad (1)$$

Each target row of the coefficients is estimated by ridge regression on its road-candidate sources only (topology-aware VAR):

$$\hat{A}_{1:3}[i, :] = \arg\min_{a} \sum_{t} \Big(x_{t,i} - \sum_{k=1}^{3} \sum_{j \in \mathcal{C}_i} a_{k,j} \, x_{t-k,j} \Big)^2 + \alpha \, \|a\|_2^2, \qquad \alpha = 10^6 \qquad (2)$$

where $\mathcal{C}_i$ is the road-candidate source set of sensor i (its road neighbours and itself). Only observed values enter. The coefficient matrices are non-zero for 1.2% of the pairs (about 25 sources per target). The residuals are $\varepsilon_{t,i} = x_{t,i} - \sum_k \sum_j A_k[i, j] \, x_{t-k,j}$. **Outputs:** A_1, A_2, A_3 and ε. Only training rows are used.

Step 2. Traffic correlation (the prior source)

The shared daily rhythm makes almost every pair of raw series strongly correlated (median $|r|$ = 0.84 on the VAR candidates), so the time-of-day profile is removed first. With $\phi(t) = t \bmod 288$ the time-of-day index:

$$\tilde{X}_{t,i} = X_{t,i} - \bar{X}_{\phi(t),i}, \qquad \bar{X}_{\phi,i} = \frac{\sum_{t:\phi(t)=\phi} m_{t,i} X_{t,i}}{\sum_{t:\phi(t)=\phi} m_{t,i}} \qquad (3)$$

Pearson correlation uses only the rows in which both sensors were observed (pairwise-complete):

$$r_{jk} = \frac{\sum_t m_{t,j} m_{t,k}\, \tilde{z}_{t,j}\, \tilde{z}_{t,k}}{\sum_t m_{t,j} m_{t,k}}, \qquad \tilde{z}_{t,i} = \frac{\tilde{X}_{t,i} - \mu_i}{\sigma_i} \qquad (4)$$

where μ_i, σ_i are the mean and standard deviation of $\tilde{X}_{\cdot,i}$ over its observed rows. Pairs with fewer than 1,000 shared rows get $r_{jk} = 0$. R is symmetric, with unit diagonal. **Output:** R.

Step 3. Prior inclusion probability

The absolute correlation is mapped linearly into a bounded prior probability. A *candidate pair* is a pair of sensors that share at least one causal unit ($\exists i : \; j, k \in U_i$); PEM only estimates these pairs.

$$\eta_{jk} = \begin{cases} 0.05 + 0.90\,|r_{jk}| & (j,k) \text{ candidate pair} \\ 0.5 & \text{otherwise (not estimated)} \end{cases} \qquad \eta_{jk} = \eta_{kj}, \;\; \eta_{jj} = 0 \qquad (5)$$

Output: η (138,548 candidate entries; mean 0.51; range 0.05 to 0.95). The uniform baseline sets $\eta_{jk} = 0.5$ for every pair. The shuffle control randomly permutes the η values among the candidate pairs (seed 42), keeping their distribution and symmetry.

Step 4. Causal units

$$U_i = \{i\} \cup \mathcal{N}_{\text{in}}(i) \cup \mathcal{N}_{\text{out}}(i), \qquad i = 1, \ldots, N \qquad (6)$$

Each unit is a small road neighbourhood, so PEM (cost growing quickly with size) runs on small problems. There are 2,056 units, one per sensor.

Step 5. PEM per unit (the probabilistic edge estimator)

5a. Unit data. Residuals are standardised per sensor over the training rows, then only the rows in which every sensor of the unit has a residual are kept:

$$z_{t,i} = \frac{\varepsilon_{t,i} - \bar{\varepsilon}_i}{s_i}, \qquad S_U = \frac{1}{n} \sum_{t \in \mathcal{T}_U} z_{t,U}\, z_{t,U}^{\top} \qquad (7)$$

where $\mathcal{T}_U$ is the set of complete rows of unit U and $n = |\mathcal{T}_U|$. The residuals are used *only* here, to estimate the variance structure.

5b. Model. The standardised residuals are modelled as Gaussian with precision matrix $\Omega = \Sigma^{-1}$. Each off-diagonal entry has a spike-and-slab Laplace prior whose slab indicator γ_{jk} has the prior inclusion probability from step 3 as its success probability:

$$\Omega_{jk} \mid \gamma_{jk} = 0 \sim \text{Laplace}(\nu_0), \quad \Omega_{jk} \mid \gamma_{jk} = 1 \sim \text{Laplace}(\nu_1), \quad \gamma_{jk} \sim \text{Bernoulli}(\eta_{jk}) \qquad (8)$$

$$f_q(w) = \frac{1}{2\nu_q} \exp\!\left(-\frac{|w|}{\nu_q}\right), \qquad \nu_1 > \nu_0 > 0, \qquad \nu_0 = 0.01,\; \nu_1 = 0.2 \qquad (9)$$

5c. MAP estimate. The precision matrix minimises the penalised negative log-likelihood, over positive-definite matrices (L-BFGS-B):

$$\hat{\Omega}_U = \arg\min_{\Omega \succ 0} \; \frac{n}{2}\Big[\operatorname{tr}(S_U\Omega) - \log\det\Omega\Big] + \tau\,\operatorname{tr}(\Omega) - \sum_{j<k} \log\Big[(1-\eta_{jk})f_0(\Omega_{jk}) + \eta_{jk} f_1(\Omega_{jk})\Big]$$

with $\tau = 0.01$. The first term fits the data, the second keeps the diagonal bounded, and the third is the correlation-informed sparsity penalty: a pair with a larger η_{jk} is penalised less for having a large precision entry.

5d. Posterior inclusion probability. At the estimate, the probability that an entry belongs to the slab is

$$P^U_{jk} = \frac{\eta_{jk}\, f_1(\hat{\Omega}_{jk})}{\eta_{jk}\, f_1(\hat{\Omega}_{jk}) + (1-\eta_{jk})\, f_0(\hat{\Omega}_{jk})} \qquad (11)$$

which equals $\operatorname{logit} P^U_{jk} = \operatorname{logit}\eta_{jk} + \log\frac{\nu_0}{\nu_1} + |\hat{\Omega}_{jk}|\big(\frac{1}{\nu_0} - \frac{1}{\nu_1}\big)$: the prior shifts the log-odds, the data evidence enters through $|\hat{\Omega}_{jk}|$.

5e. Combining units. A pair can occur in several units; the largest probability is kept:

$$P_{jk} = \max_{U \ni j,k} P^U_{jk}, \qquad P_{jk} = P_{kj}, \qquad P_{jk} = 0 \text{ if no unit contains both} \qquad (12)$$

Output: P (symmetric, $[0,1]$). PEM also returns an adjacency by thresholding at 0.5; E8 uses only P. P is symmetric, so it cannot give a direction.

Step 6. VAR impulse responses (direction and size)

The effect of source j on target i after s steps follows from the VAR coefficients:

$$\Psi_0 = I, \qquad \Psi_s = \sum_{k=1}^{\min(s,3)} A_k\, \Psi_{s-k}, \qquad h \in \{1,3,6,12\} \qquad (13)$$

so $\Psi_1 = A_1$ (checked in the code). Ψ_h is directed and includes indirect (multi-step) effects. **Output:** $\Psi_1, \Psi_3, \Psi_6, \Psi_{12}$.

Step 7. Combining VAR and PEM, then top-K

$$S_h[i,j] = P_{ij}\,\big|\Psi_h[i,j]\big| \quad (i \neq j), \qquad S_h[i,i] = 0 \qquad (14)$$

$$G_h[i,j] = \begin{cases} S_h[i,j] & j \text{ is among the } K = 25 \text{ largest entries of } S_h[i,:] \\ 0 & \text{otherwise} \end{cases} \qquad (15)$$

The product keeps the direction and magnitude from VAR and down-weights edges that PEM finds unlikely. Because P is zero for pairs sharing no unit, indirect pairs outside the road neighbourhood are dropped. Each target has at most 25 incoming sources. No LiNGAM and no B_0 is used. **Output:** G_h, stored as $G[\text{target}, \text{source}]$.

Step 8. Forecasting with A3T-GCN (unchanged)

$$\hat{A}_h = D^{-1/2}\,(G_h + I)\,D^{-1/2}, \qquad D_{jj} = 1 + \sum_i G_h[i,j] \qquad (16)$$

The graph is transposed once for the model orientation. The input is the last 12 steps of flow; the forecast is made at horizon h using the graph of horizon h. Training: 20 epochs, seed 42 for the first pass, the best-validation epoch is kept, and the test set is evaluated once. Finalists are repeated with at least 3 seeds. The

model, optimiser, learning rate, batch size and early-stopping rule are not changed. The normalisation shown is taken from the earlier E5 documentation.

6. Experiments and controls

Run	Prior η	Edge weight	Purpose
uniform	0.5 for all pairs	$P\,\lvert\Psi_h\rvert$	PEM without any prior information
flow (proposed)	$0.05 + 0.90\lvert r_{jk}\rvert$	$P\,\lvert\Psi_h\rvert$	correlation-informed prior
flow_shuffle	same values as flow, randomly reassigned (seed 42)	$P\,\lvert\Psi_h\rvert$	control: varying prior without correlation information
E8V	none (no PEM)	$\lvert\Psi_h\rvert$	control: VAR only; E8 must beat it for PEM to count as useful

Reading the results: if *flow* beats *flow_shuffle*, the correlation information helps; if both beat *uniform* but not each other, any varying prior helps; if *flow* does not beat E8V, PEM does not help. All comparisons use the validation set for selection and the test set once, with at least 3 seeds for finalists.

7. Fixed settings

Setting	Value
Dataset / split	LargeST-GBA 2021, 5-min flow; chronological; training rows 0 to 84,096, validation rows 84,096 to 94,608; test once
Sensors	2,056 (duplicate filter)
VAR	lag $p = 3$, ridge $\alpha = 10^6$, road-candidate sources, observed values only
Correlation	Pearson, pairwise-complete, time-of-day mean removed (period 288), at least 1,000 shared rows
Prior	$\eta = 0.05 + 0.90\lvert r\rvert$ on candidate pairs, 0.5 elsewhere, symmetric
PEM	$\nu_0 = 0.01,\ \nu_1 = 0.2,\ \tau = 0.01$, threshold 0.5, standardised residuals, L-BFGS-B (SLSQP retry if needed), complete rows per unit
Horizons / graph	$h \in \{1, 3, 6, 12\}$, top-$K = 25$ sources per target, $G[\text{target}, \text{source}]$
A3T-GCN	unchanged; 20 epochs; seed 42 first pass; best-validation epoch; test once; at least 3 seeds for finalists

The values $\nu_0 = 0.01, \nu_1 = 0.2$ replace the earlier $0.0017,\ 1$, which leaves P at 0 or 1 for almost every pair (85% above 0.99, 12% below 0.01), so no prior could matter. The change was made after observing this saturation and is reported as post-hoc.

8. Checks built into the code

- $\Psi_1 = A_1$ (orientation and recursion); P symmetric and within [0,1]; all VAR coefficients finite.
- Prior symmetric and within [0.05, 0.95]; correlation matrix symmetric; only training rows enter the VAR, correlation, prior and PEM.
- Graph diagonal zero; at most 25 incoming sources per target; every selected edge lies where $P > 0$; the PEM result and VAR run share sensors and training split.
- No LiNGAM module is imported and no B_0 file is opened (reported at the end of the build).

- Unit tests: 7 for the correlation and prior module (Pearson, Spearman, partial, mapping) and one for the flow correlation (detrending and missing values).

9. Check done so far (local, training data only)

PEM was run on a reproducible sample of 60 units (seed 42), once with the uniform prior and once with the flow prior, at $\nu_0 = 0.01, \nu_1 = 0.2$. All 60 units converged.

Measure (flow prior compared with uniform prior)	Value
Entries of P changed by more than 0.01	54.6%
Entries changed by more than 0.10	14.3%
Entries crossing the 0.5 threshold	4.5%
Mean absolute change of P	0.047
P strictly between 0.05 and 0.95 (graded)	44.4% (uniform baseline: 40.5%)
P above 0.99 / below 0.01	35.0% / 0.5%

The prior therefore changes the probabilities moderately and does not saturate them. This is a graph-level check on a 60-unit slice; it says nothing about forecasting accuracy.

10. Assumptions and limitations

- No claim that E8 forecasts better is made before the A3T-GCN runs. The prior changes P moderately; whether this helps forecasting is open.
- P is symmetric and undirected, so it only weights edges. Direction and size come from VAR; pairs sharing no road unit get $P = 0$.
- The correlation of the detrended flow partly reflects spatial proximity; it is association, not causation.
- P is a plug-in slab probability computed at the MAP estimate, not the posterior probability that an edge is real.
- The VAR companion matrix has an eigenvalue of 1.05 (> 1), so $|\Psi_h|$ grows with the horizon; this affects the long horizons for all VAR-based graphs.
- The ν change is post-hoc. Dynamic re-estimation of VAR and PEM after each A3T-GCN result is a later experiment and is not part of E8.